

12345678

A

power_input

File: power_input.kicad_sch

boost_charger

File: boost_charger.kicad_sch

cap_bank

File: cap_bank.kicad_sch

pulse_switch

File: pulse_switch.kicad_sch

flyback

File: flyback.kicad_sch

sense_gates

File: sense_gates.kicad_sch

mcu_status

File: mcu_status.kicad_sch

OmniMarble

Sheet: /
File: omnimarble-driver.kicad_sch

Title: OmniMarble 60V SELV coilgun driver

Size: A3Date:

KiCad E.D.A. 10.0.4Id: 8/8

B

C

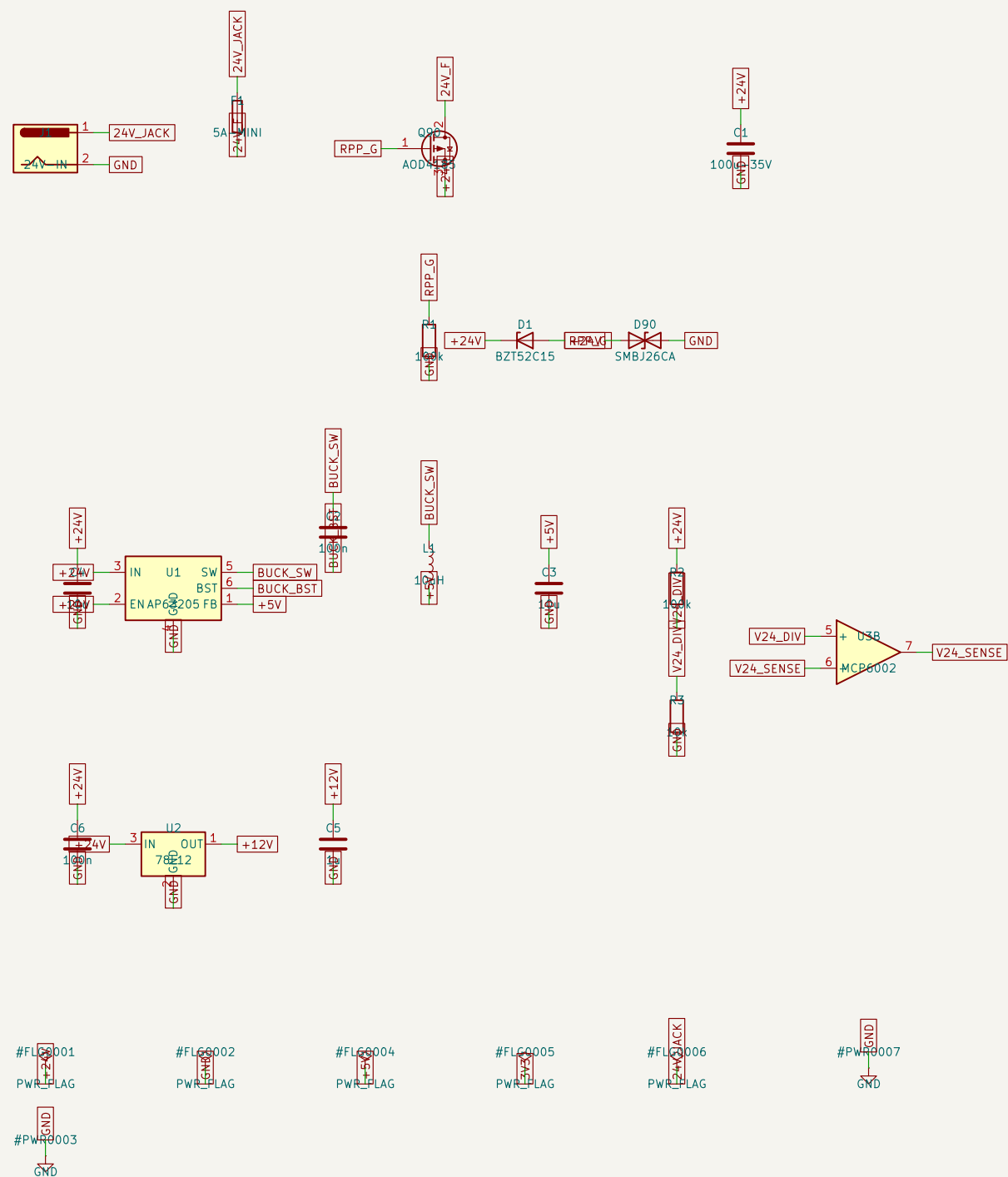
D

E

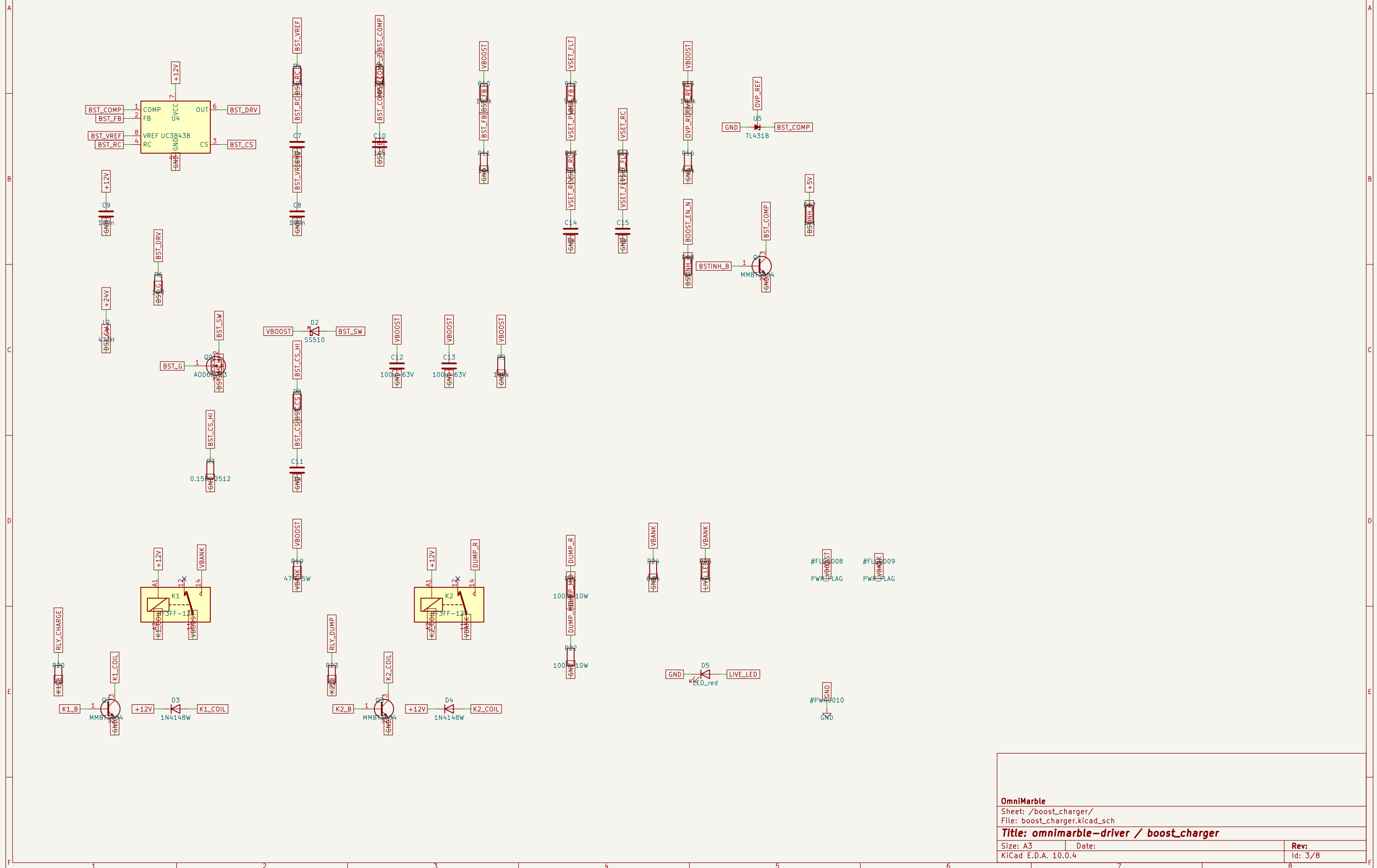
F

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out, protection, 5V buck, 12V rail

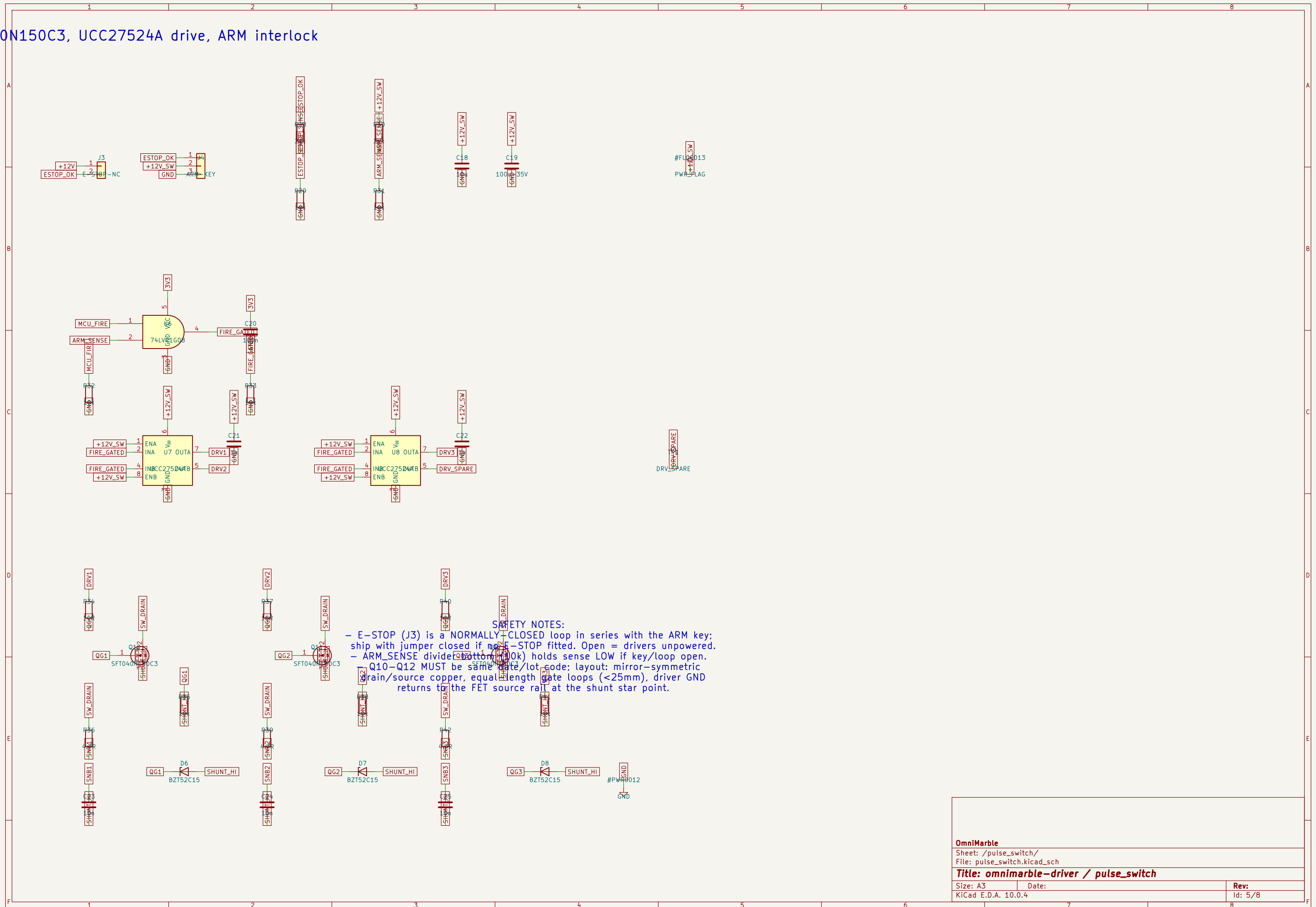


5V operating (63V-rated bank), OVP, relays, bleed

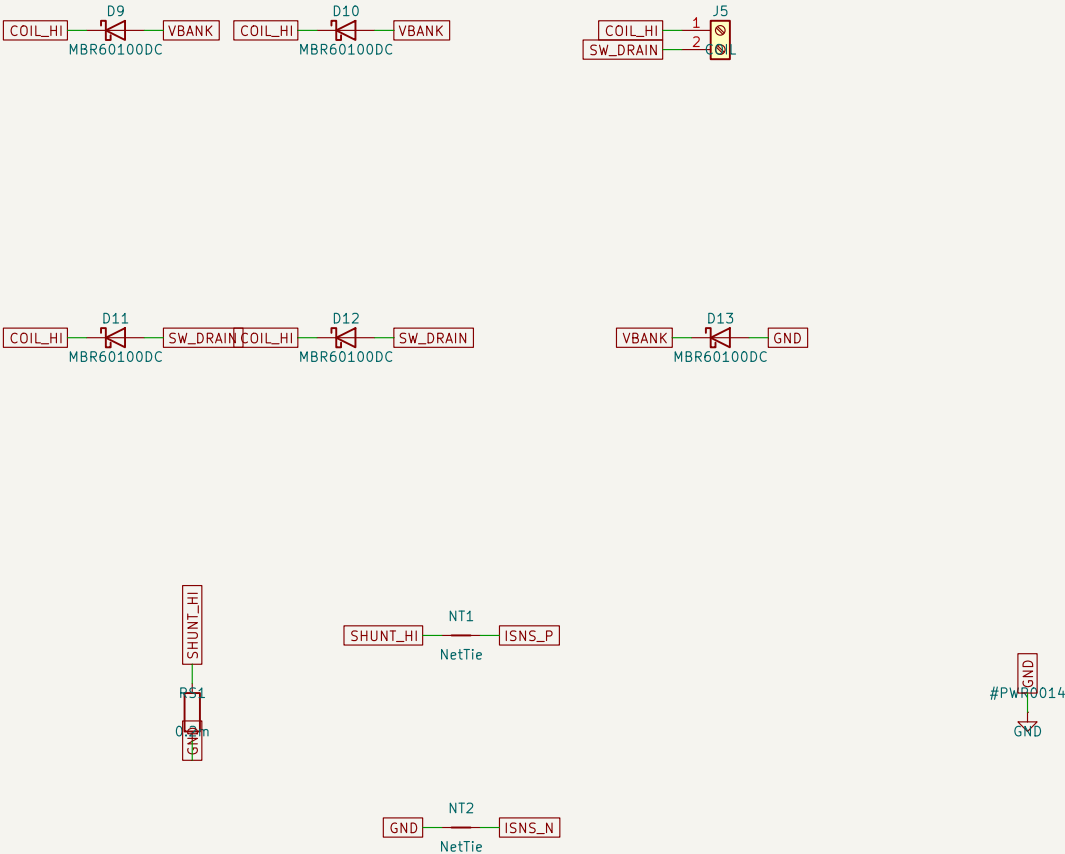


2200µF/63V snap-in (populate per bank table)

T040N150C3, UCC27524A drive, ARM interlock



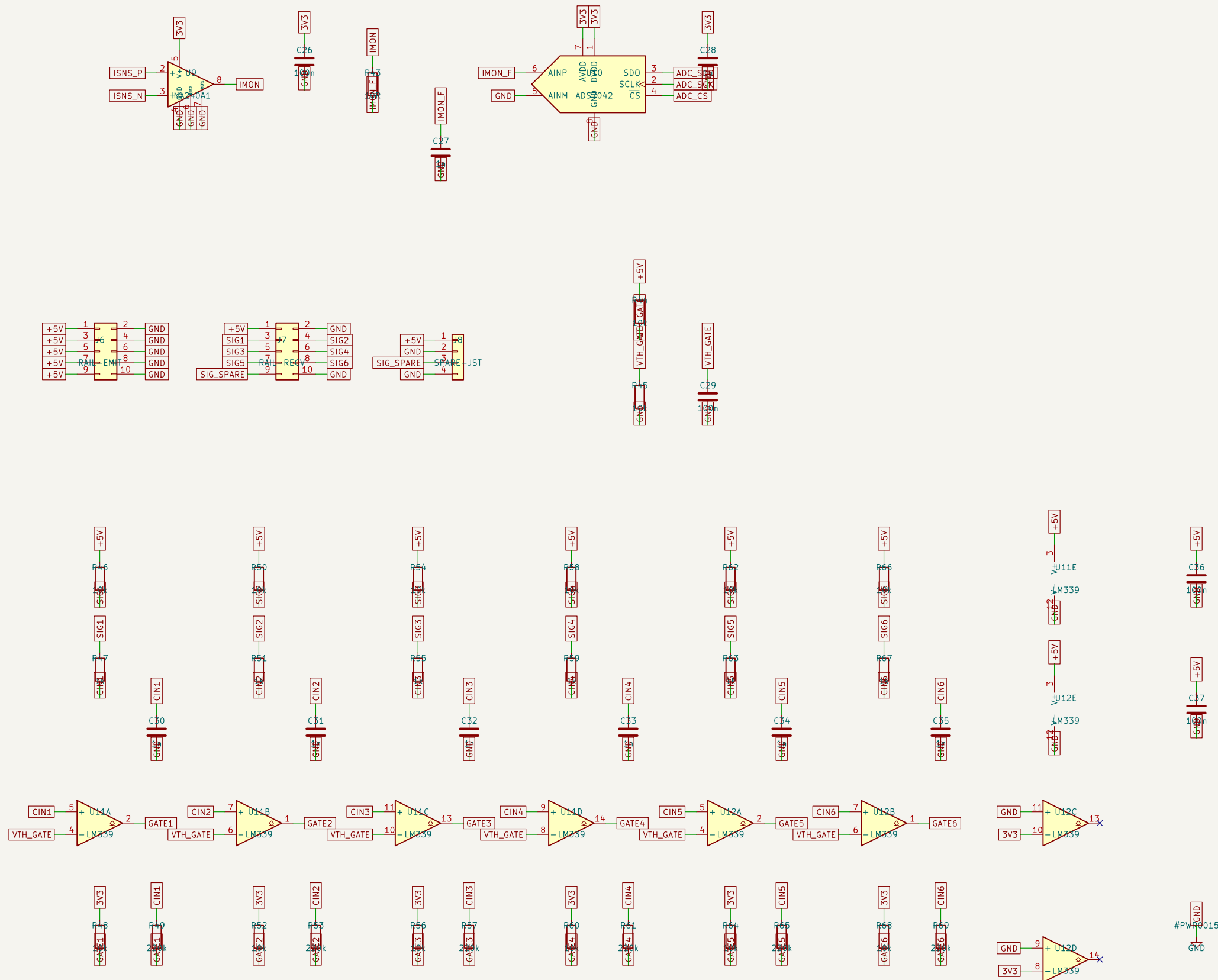
blocking -> COIL -> switch -> shunt -> GND



- validated envelope, swept in calcs.py at 11000uF / 55V:
>= 1 uH AND R_total >= 90 mOhm
coil DC resistance + leads/lugs; keep leads short)
fault: 30T demo coil, 12.406 uH, 110 mOhm total
se can exceed the 600A switch design point – DO NOT FIRE.

OmniMarble		
Sheet: /flyback/ File: flyback.kicad_sch		
Title: omnimarble-driver / flyback		
Size: A3	Date:	Rev:
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42 current capture; 6x IR gate comparators



IR GATE TRUTH TABLE:

T -> phototransistor ON -> SIGx -0V -> comparator OUT LOW
 -> phototransistor OFF -> SIGx -5V -> comparator OUT HIGH
 GH = marble in beam. Firmware timestamps the RISING edge.
 open-collector outputs pulled to 3V3: Pico-safe levels)

OmniMarble	
Sheet: /sense_gates/ File: sense_gates.kicad_sch	
Title: omnimarble-driver / sense_gates	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Rev: Id: 7/8

0), MCU-agnostic breakout, relay/status drive

